

Engineering Mathematics I

Learner Notes

Engineering calculations supporting structural analysis, survey principles, measurement, and costing.

Learning outcome	Focus
Apply algebraic methods	Applied mathematics in workplace and training contexts.
Use trigonometry in building works	Applied mathematics in workplace and training contexts.
Apply mensuration in construction	Applied mathematics in workplace and training contexts.

1. Topic Overview

This topic develops mathematical skills required in Building Technology. Learners should show clear method, correct units, and technical interpretation.

2. Key Formulas

$$y = mx + c$$

$$\tan \theta = \frac{\text{rise}}{\text{run}}$$

$$V = \frac{1}{3}Ah$$

$$C = \pi d$$

3. Worked-Example Method

Step 1: identify the given data. Step 2: select the correct formula. Step 3: substitute values carefully. Step 4: compute. Step 5: interpret the answer with correct units.

$$y = mx + c$$

4. Practice Questions

1. Calculate roof pitch angle.

2. Find volume of excavation.
3. Interpret a straight-line graph from site data.